Big Bang beginning with, 32–35, 40n
black hole and, 93, 94n
bounce model and, 170
Conformal Cyclic Cosmology and, 176
cyclic universes and, 66, 175
ekpyrotic mode and, 174
Solar System
Andromeda Galaxy–Milky Way collision in, 51–52, 62
definitions and measurements of distance in, 116–17
distance ladder in, 117, 119 (fig.)
vacuum decay in, 150
sound, and Doppler effect, 54, 55 (fig.)
spacetime
dark energy as property of, 81
description of, 18–19
gravity in, 164
light moving through, 20 (fig.)
spectrum pattern characteristics, in star identification, 27, 55–57
spontaneous symmetry breaking, 134
standard candle method of distance measurement, 73–74, 117–18, 119
Standard Model of particle physics, 181–83, 190, 191
Higgs boson and, 130, 134, 141, 181
physical phenomena not explained in, 161–62
vacuum decay and, 155
stars and star systems. *See also* white dwarf stars
Andromeda Galaxy–Milky Way collision and formation of, 51–52
astronomical all-sky surveys of, 113
Big Crunch impact on, 62–63, 65
Big Rip and, 113
Cepheid variable stars, 118–19, 119 (fig.), 127
Chandrasekhar Limit and, 124–25
Cosmic Dawn and beginning of, 49
cosmic expansion and, 61
distance ladder measurements and, 119 (fig.)
electron degeneracy pressure and collapse of, 122–23, 124
Epoch of Reionization and, 49
gravitational wave astronomy on, 158